Lesson 4: Add a target and detect collision

Pygame CE • S3 onwards • 50-60 min

Learning intention
Create a second object and detect when two rectangles overlap.

Success criteria

- I can create a `pygame.Rect`.
- I can detect a collision with `colliderect()`.
- I can move the target to a random position.

Key ideas

- **pygame.Rect(x, y, width, height)** stores a rectangle and has useful methods.
- **colliderect()** returns True when two rectangles overlap.
- **random.randint(a, b)** gives a whole number between a and b inclusive.

Activities

1. Add the target and draw it in a different colour.
2. Create the player's Rect each frame from x and y.
3. Move the target when it is collected.
4. Pair-test: does it ever appear partly off screen?

Code

Key changes

```
import random

target = pygame.Rect(random.randint(0, WIDTH - SIZE),
                     random.randint(0, HEIGHT - SIZE), SIZE, SIZE)

# inside the loop, after movement:
player_rect = pygame.Rect(int(x), int(y), SIZE, SIZE)
if player_rect.colliderect(target):
    target.topleft = (random.randint(0, WIDTH - SIZE),
                      random.randint(0, HEIGHT - SIZE))

pygame.draw.rect(screen, (240, 190, 60), target)
```

Complete program

```
import pygame
import random

pygame.init()

WIDTH = 800
HEIGHT = 600
SIZE = 40

screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("Collect the target")
clock = pygame.time.Clock()

x = 380.0
y = 280.0
speed = 250
target = pygame.Rect(random.randint(0, WIDTH - SIZE),
```

```
                    random.randint(0, HEIGHT - SIZE), SIZE, SIZE)
running = True

while running:
    dt = clock.tick(60) / 1000

    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    x += (keys[pygame.K_RIGHT] - keys[pygame.K_LEFT]) * speed * dt
    y += (keys[pygame.K_DOWN] - keys[pygame.K_UP]) * speed * dt
    x = max(0, min(x, WIDTH - SIZE))
    y = max(0, min(y, HEIGHT - SIZE))

    player_rect = pygame.Rect(int(x), int(y), SIZE, SIZE)
    if player_rect.colliderect(target):
        target.topleft = (random.randint(0, WIDTH - SIZE),
                          random.randint(0, HEIGHT - SIZE))

    screen.fill((30, 30, 40))
    pygame.draw.rect(screen, (240, 190, 60), target)
    pygame.draw.rect(screen, (80, 200, 120), player_rect)
    pygame.display.flip()

pygame.quit()
```

Checkpoint

The target jumps to a new random position each time the player touches it.

Support and stretch

- **Support:** Provide the Rect code and ask learners to place it in the right part of the loop.
- **Stretch:** Make the target a different size, or make sure it never appears under the player.

Reflect

Why do we check for collisions every frame rather than just once?

Save your work
Save this version as `lesson4.py` before you start the next lesson.